Workshop

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2020020

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GUI.

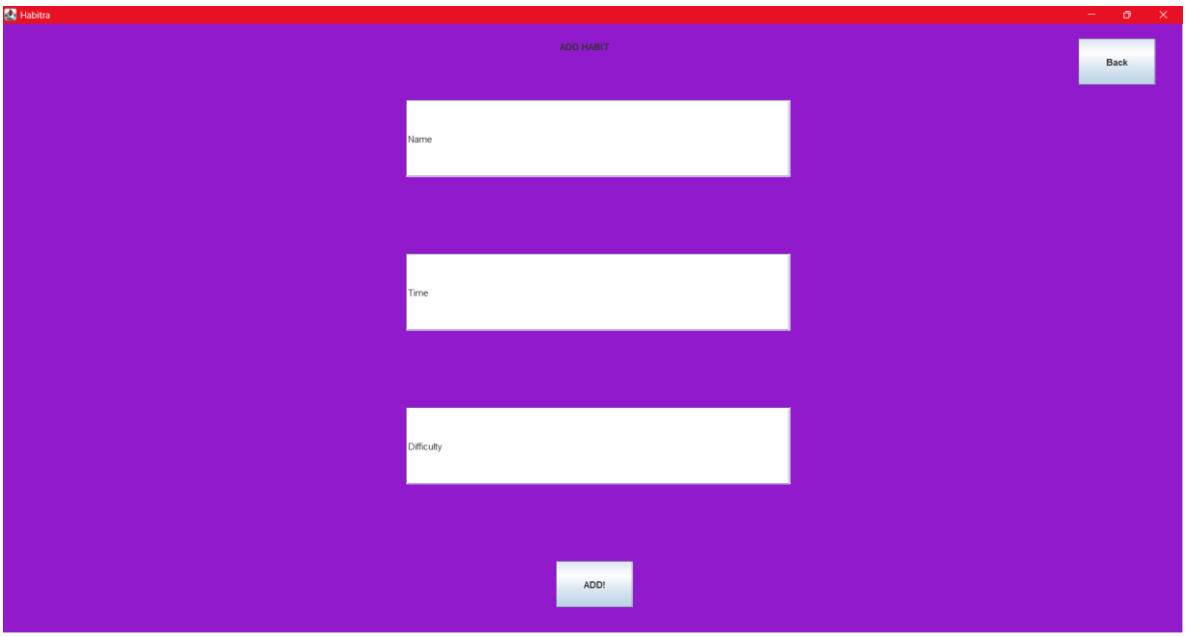
Objectives

Generate a Graphical User Interface (GUI) for our application Habitra, designing early prototypes that try to be as faithful as possible to the mockups made at the early stages of development.

Development

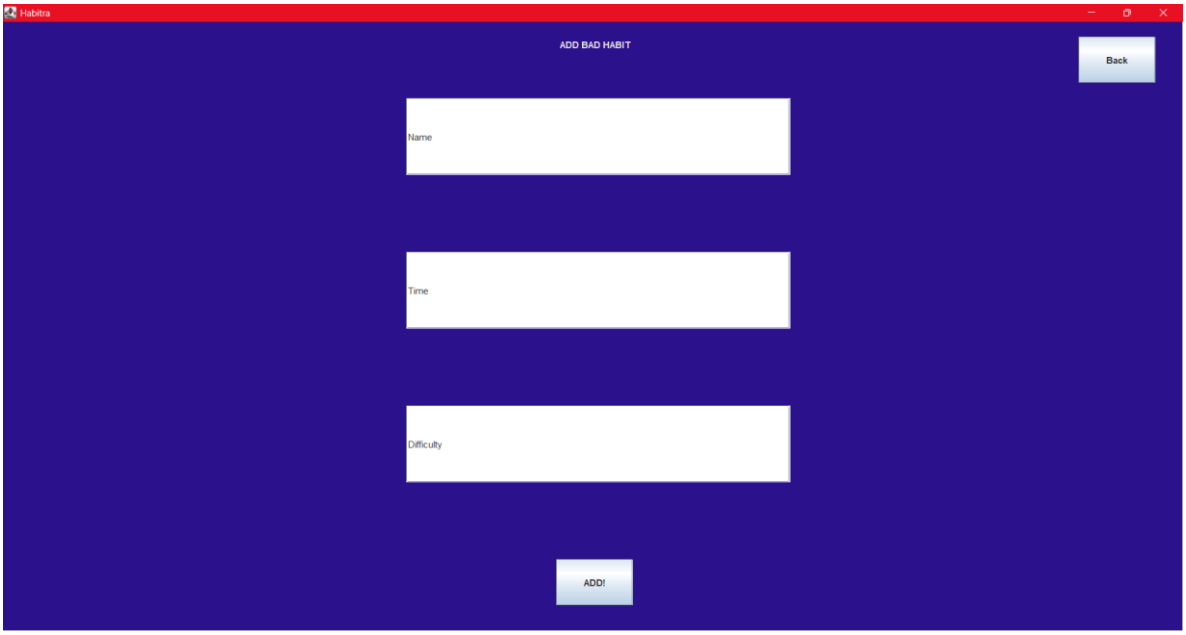
Prototype for user interface:

Main Menu with three “add” buttons



Add Habit screen

Add Task Screen



Add Bad Habit Screen

These are early prototypes on how the application should look and feel like.

The user selects one of three options, provides the necessary data, and then can add a new item to the list of items, or cancel adding by clicking on the “back” button.

Progress in code:

```java
import javax.swing.*;
import java.awt.*;

public class App{
    //Declares objects as attributes, this helps to simply initiate them instead of declaring them every time we want one of these
    public static JPanel panel, newPanel;
    public static JFrame frame;
    public static JButton addH, addT, addB, back, add;
    public static JTextField name, deadline, time, diff;
    public static JLabel label;
    public static void main(String[] args) {
        startFrame();
        startButtons();

        panel.setLayout(null);
        panel.add(addH);
        panel.add(addT);
        panel.add(addB);

        panel.setBackground(Color.decode("#472287"));


    }
```

Code that initializes the frame and first panel

```java
/**Redirects to the corresponfing panel */
   private static void event(int id){
       panel.setVisible(false);
       newPanel = new JPanel();
       newPanel.setBounds(-400, 0, 1920, 1080);
       switch (id) {
           case 1:
               newPanel.setBackground(Color.decode("#911bcc"));
               AddHabit();
               break;
           case 2:
               newPanel.setBackground(Color.decode("#8c1176"));
               AddTask();
               break;
           case 3:
               newPanel.setBackground(Color.decode("#2c118c"));
               AddBadHabit();
               break;
           default:
               newPanel.setBackground(Color.RED);
               break;
       }

       newPanel.setVisible(true);
       newPanel.setLayout(null);
       frame.add(newPanel);
       //Creates a button that goes back to the main panel
       back= new JButton("Back");
       back.setBounds(1400, 20, 100, 60);
       back.addActionListener(e->{
           newPanel.setVisible(false);
           panel.setVisible(true);
       });

       newPanel.add(back);

       frame.revalidate();
       frame.repaint
```

Method that handles redirecting from the main panel

```java
private static void AddHabit(){
        label = new JLabel("ADD HABIT");
        label.setBounds(725, 20, 100, 20);
        label.setVisible(true);

        name = new JTextField("Name");
        name.setBounds(525, 100, 500, 100);
        name.setVisible(true);

        time= new JTextField("Time");
        time.setBounds(525, 300, 500, 100);
        time.setVisible(true);

        diff= new JTextField("Difficulty");
        diff.setBounds(525, 500, 500, 100);
        diff.setVisible(true);

        addButton();

        newPanel.add(label);
        newPanel.add(name);
        newPanel.add(time);
        newPanel.add(diff);

    }
```

Code that generates the “Add Habit” screen textboxes

```java
private static void AddTask(){
        label = new JLabel("ADD TASK");
        label.setBounds(725, 20, 100, 20);
        label.setVisible(true);

        name = new JTextField("Name");
        name.setBounds(525, 100, 500, 100);
        name.setVisible(true);

        deadline= new JTextField("Deadline");
        deadline.setBounds(525, 300, 500, 100);
        deadline.setVisible(true);

        diff= new JTextField("Difficulty");
        diff.setBounds(525, 500, 500, 100);
        diff.setVisible(true);

        addButton();

        newPanel.add(label);
        newPanel.add(name);
        newPanel.add(deadline);
        newPanel.add(diff);
    }
```

Code that generates the “Add Task” screen textboxes

```java
private static void AddBadHabit(){
        label = new JLabel("ADD BAD HABIT");
        label.setForeground(Color.decode("#f0e9e9"));
        label.setBounds(725, 20, 100, 20);
        label.setVisible(true);

        name = new JTextField("Name");
        name.setBounds(525, 100, 500, 100);
        name.setVisible(true);

        time= new JTextField("Time");
        time.setBounds(525, 300, 500, 100);
        time.setVisible(true);

        diff= new JTextField("Difficulty");
        diff.setBounds(525, 500, 500, 100);
        diff.setVisible(true);

        addButton();

        newPanel.add(label);
        newPanel.add(name);
        newPanel.add(time);
        newPanel.add(diff);
    }
```

Code that generates the “Add Bad Habit” screen textboxes

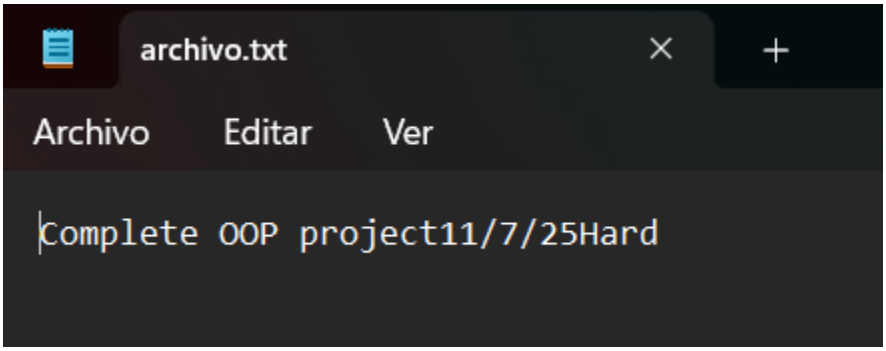
```java
/**Starts the frame with a panel */
    private static void startFrame(){
        frame = new JFrame("Habitra");
        panel = new JPanel();
        frame.setSize(1920,1080);
        frame.setVisible(true);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.add(panel);
    }
    /**Starts the buttons */
    private static void startButtons(){
        addH= new JButton("AddHabit");
        addH.setBounds(30, 500, 100, 60);
        addH.addActionListener(e -> {
            event(1);
        });

        addT= new JButton("AddTask");
        addT.setBounds(30, 600, 100, 60);
        addT.addActionListener(e -> {
            event(2);
        });

        addB= new JButton("AddBadHabit");
        addB.setBounds(30, 700, 130, 60);
        addB.addActionListener(e -> {
            event(3);
        });
    }
    /**Creates an "add" button */
    private static void addButton(){
        add= new JButton("ADD!");
        add.setBounds(720, 700, 100, 60);
        add.setVisible(true);
        add.addActionListener(e->{
            newPanel.setVisible(false);
            panel.setVisible(true);
        });

        newPanel.add(add);
```

Methods that initialize the frame, main screen buttons and the add button

File Usage and Data Persistence:

"Add Task" text fields with some data on it.

When the user adds a new habit or task...



File text with user input.

...the app recollects this information and writes it in a text file for later use.

```java
/**Creates an "add" button for habits */
    private static void addButtonHabit(){
        addButton();
        add.addActionListener(e->{
            try{
                writer= new FileWriter("archivo.txt");
                writer.write(name.getText());
                writer.write(time.getText());
                writer.write(diff.getText());
                writer.close();
            }
            catch(IOException x){
                System.out.println("Impossible to write on file: "+x);
            }
            newPanel.setVisible(false);
            panel.setVisible(true);
        });

        newPanel.add(add);
    }

    /**Creates an "add" button for tasks*/
    private static void addButtonTask(){
        addButton();

        add.addActionListener(e->{
            try{
                writer= new FileWriter("archivo.txt");
                writer.write(name.getText());
                writer.write(time.getText());
                writer.write(diff.getText());
                writer.close();
            }
            catch(IOException x){
                System.out.println("Impossible to write on file: "+x);
            }
            newPanel.setVisible(false);
            panel.setVisible(true);
        });

        newPanel.add(add);
    }
}
```

Code that handles writing on file.